Tragedy of The Commons and The Impact of Human Activities on the Environment

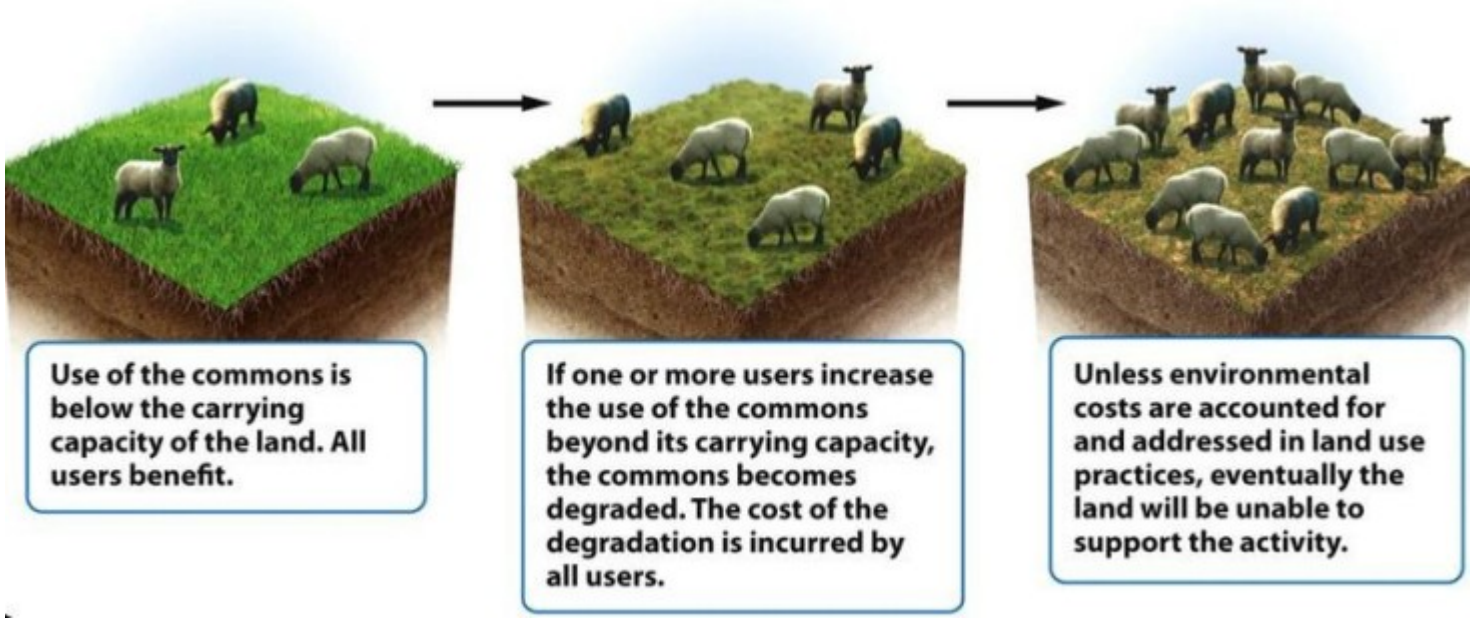
WI1103 Introduction of Sustainability Principles Week-2)

Semester I 2026/2027



Pengantar Konsep *Tragedy of The Commons*

"Picture a pasture that is open to all. Each herdsman will try to keep as many cattle as possible on the commons…the inherent logic of the commons remorselessly generates tragedy." ---Garrett Hardin, *The Tragedy of the Commons* 1968



"the commons" **is a resource that is "open to all" (open access)**

Commons vs open-access resource

- The commons proposed by Hardin is debated, because according to the literature, **commons refers to shared ownership**

- The difference between commons and open-access resources emphasizes that **tragedy can occur in open-access resources that are not well managed**.

	The commons	Open-access resource
Ownership	Joint ownership	There are no ownership rights or possession.
Definition of Community	A group that is defined and restricted more narrowly with certain boundaries between members and non-members.	The community at large
Level of exclusion	Exclusion for non-members	None

Open-access resources

- Some renewable resources, known as **open-access resources**, are not owned by anyone and can be used by almost anyone.
- Examples include the atmosphere and the open sea along with its fish.
- Another example that is not entirely open but is often used collectively includes grasslands, forests, rivers, and underground water bodies (aquifers).

Does the tragedy lie on the commons?

Contoh 1: Lobster fisheries in Maine, UK, have policy mechanisms among local communities to protect water resources as well as lobsters and to avoid overfishing of lobsters.



Contoh 2: Customary forest Barambang Katute

The customary forest area is managed by the Barambang Katute indigenous community, which has mechanisms in place to protect this forest area. There are sacred areas where mining activities are not allowed.

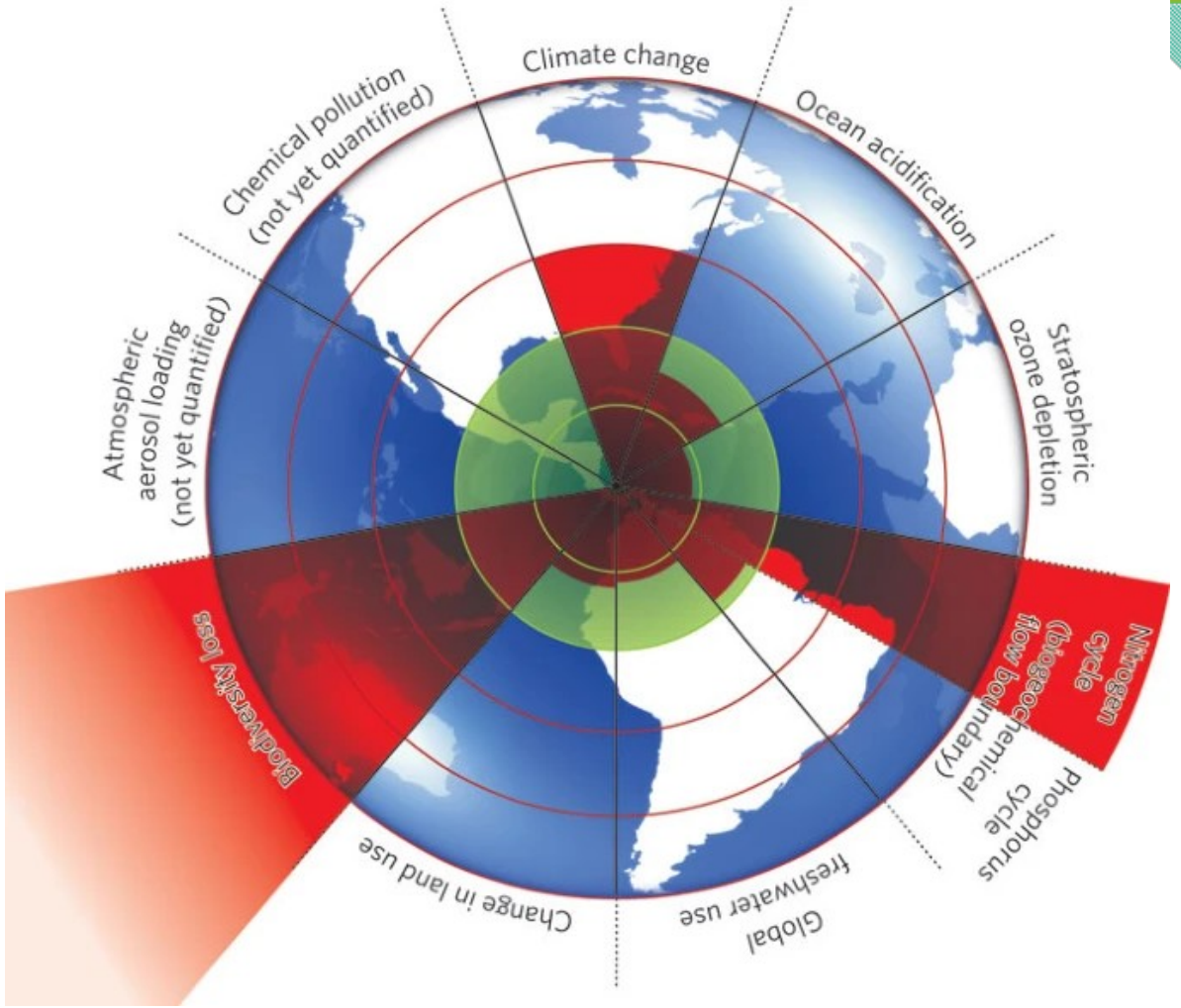


Planetary Boundaries

The planetary boundaries framework draws upon Earth system science. It identifies nine processes that are critical for maintaining the stability and resilience of Earth system as a whole.

The planetary boundaries framework delineates the biophysical and biochemical systems and processes known to regulate the state of the planet within ranges that are historically known and scientifically likely to maintain Earth system stability and life-support systems conducive to the human welfare and societal development.

The planetary boundaries framework formulates limits to the impact of the anthroposphere on Earth system by identifying a scientifically based safe operating space for humanity that can safeguard both Earth's interglacial state and its resilience.



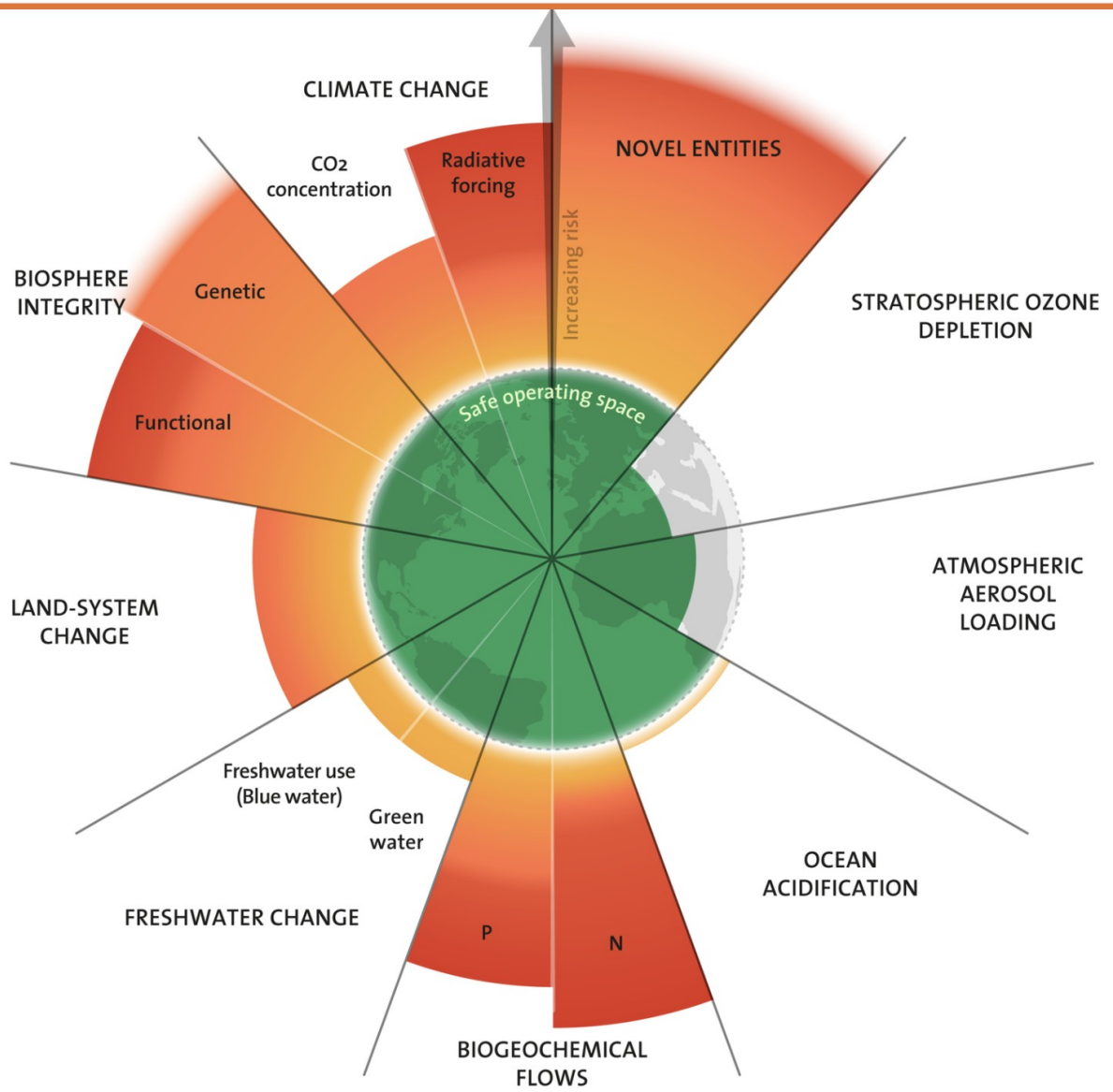
https://www.nature.com/articles/461472a

Planetary Boundaries

7 out of 9 planetary boundaries have been transgressed

indicating that the Earth is now outside the safe operating space for humanity.

https://www.stockholmresilience.org/news--events/general-news/2025-09-24-seven-of-nine-planetary-boundaries-now-breached.html



The 2025 update to the Planetary boundaries. Licensed under CC BY-NC-ND 3.0. Credit: "Azote for Stockholm Resilience Centre, based on analysis in Sakschewski and Caesar et al. 2025".

Changes in Land Use

Large-scale land conversion for agriculture, urbanization, and deforestation has disrupted natural ecosystems, reduced biodiversity, and affected carbon and water cycles.



Global Water Cycle

Changes in water use, such as large-scale irrigation and river diversion, have altered the hydrological cycle, causing droughts, floods, and a decline in water quality.

Climate Change

Greenhouse gas emissions, particularly from the burning of fossil fuels, have increased CO2 concentrations in the atmosphere, causing global warming and climate change.

Loss of biodiversity

Human activities, such as habitat destruction, pollution, and resource exploitation, have caused species extinction at a very high rate, disrupting ecosystem functions.

Biogeochemical flow:

The excessive use of synthetic fertilizers in agriculture has resulted in the excessive flow of nitrogen and phosphorus into the ecosystem, causing eutrophication and disruptions in the natural cycle.



Eutrofikasi

Chemical pollution:

The use of hazardous chemicals, including pesticides, heavy metals, and industrial chemicals, has polluted the environment, affecting the health of humans and animals.



Chemical Polution

Natural Capital Degradation

Major Human Impacts on Terrestrial Ecosystems



Deserts	Grasslands	Forests	Mountains
Large desert cities	Conversion to cropland	Clearing for agriculture, livestock grazing, timber, and urban development	Agriculture
Destruction of soil and underground habitat by off-road vehicles	Release of CO_2 to atmosphere from burning grassland	Conversion of diverse forests to tree plantations	Timber and mineral extraction
Depletion of groundwater	Overgrazing by livestock	Damage from off-road vehicles	Hydroelectric dams and reservoirs
Land disturbance and pollution from mineral extraction	Oil production and off-road vehicles in arctic tundra	Pollution of forest streams	Air pollution blowing in from urban areas and power plants
			Soil damage from off-road vehicles

Miller and Spoolman, 2019

Natural Capital Degradation

Major Human Impacts on Marine Ecosystems and Coral Reefs

Marine Ecosystems

Coral Reefs



Half of coastal wetlands lost to agriculture and urban development

Over one-fifth of mangrove forests lost to agriculture, aquaculture, and development

Beaches eroding due to development and rising sea levels

Ocean-bottom habitats degraded by dredging and trawler fishing

At least 20% of coral reefs severely damaged and 25–33% more threatened

Ocean warming

Rising ocean acidity

Rising sea levels

Soil erosion

Algae growth from fertilizer runoff

Bleaching

Increased UV exposure

Damage from anchors and from fishing and diving

Top left: travelview/Shutterstock.com. Top right: Rich Carey/Shutterstock.com. Bottom left: B-D-S Piotr Marcinski/Shutterstock.com. Bottom right: Rostislav Ageev/Shutterstock.com.

Miller and Spoolman, 2019

Discussion

- Imagine a future where two other planetary boundaries have been breached. In your opinion, what would the state of the world be like then? How might daily life change?

- Discuss how local actions, such as reducing plastic use or saving water, contribute to maintaining planetary boundaries. Are these actions enough, or do we need more global efforts?

- With knowledge of planetary boundaries, what ethical responsibilities do individuals, corporations, community, and governments have to prevent violations of those limits?

Terima kasih